HDOC Day-40 - C Programming

Menu-Driven String Operations Using User-Defined Functions and Pointers

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Batch	15 / 16
Header Files	stdio.h, string.h, ctype.h

Aim

To write a menu-driven C program that reads a student's first, middle, and last name and performs string operations using separate user-defined functions. Pointers are used wherever a function modifies data or returns results.

Operations Performed

1. Combine first, middle, and last names using strcpy() and strcat().

2. Find the length of the full name using strlen().

3. Copy the full name using strcpy().

4. Compare first and last names using strcmp().

5. Display the name in last-middle-first order.

6. Count vowels, consonants, digits, and spaces using ctype.h functions.

7. Search for a word in the full name using strstr().

8. Exit the program.

C Program

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void combineNames(const char firstName[], const char middleName[],
                  const char lastName[], char fullName[])
{
    strcpy(fullName, firstName);
    strcat(fullName, " ");
    strcat(fullName, middleName);
    strcat(fullName, " ");
    strcat(fullName, lastName);
}

void findLength(const char fullName[], size_t *length)
{
    *length = strlen(fullName);
}

void copyName(const char source[], char destination[])
{
    strcpy(destination, source);
}

void compareNames(const char firstName[], const char lastName[],
                  int *comparisonResult)
{
    *comparisonResult = strcmp(firstName, lastName);
}

void reverseNameOrder(const char firstName[], const char middleName[],
                      const char lastName[], char reverseName[])
{
    strcpy(reverseName, lastName);
    strcat(reverseName, " ");
    strcat(reverseName, middleName);
    strcat(reverseName, " ");
    strcat(reverseName, firstName);
}

void countCharacters(const char fullName[], int *vowels, int *consonants,
                     int *digits, int *spaces)
{
    int i;
    char ch;
    *vowels = *consonants = *digits = *spaces = 0;

    for (i = 0; i < (int)strlen(fullName); i++) {
        ch = (char)tolower((unsigned char)fullName[i]);
        if (isalpha((unsigned char)ch)) {
            if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
                (*vowels)++;
            else
                (*consonants)++;
        } else if (isdigit((unsigned char)ch)) {
            (*digits)++;
        } else if (isspace((unsigned char)ch)) {
            (*spaces)++;
        }
    }
}

void searchWord(const char fullName[], const char word[], int *searchResult)
{
    *searchResult = (strstr(fullName, word) != NULL);
}

int main(void)
{
    char firstName[50], middleName[50], lastName[50];
    char fullName[160] = "", copiedName[160], reverseName[160], word[50];
    int choice, comparisonResult, searchResult;
    int vowels, consonants, digits, spaces;
    size_t length;

    printf("Enter first name: ");
    scanf("%49s", firstName);
    printf("Enter middle name: ");
```

```
    scanf("%49s", middleName);
    printf("Enter last name: ");
    scanf("%49s", lastName);

    combineNames(firstName, middleName, lastName, fullName);

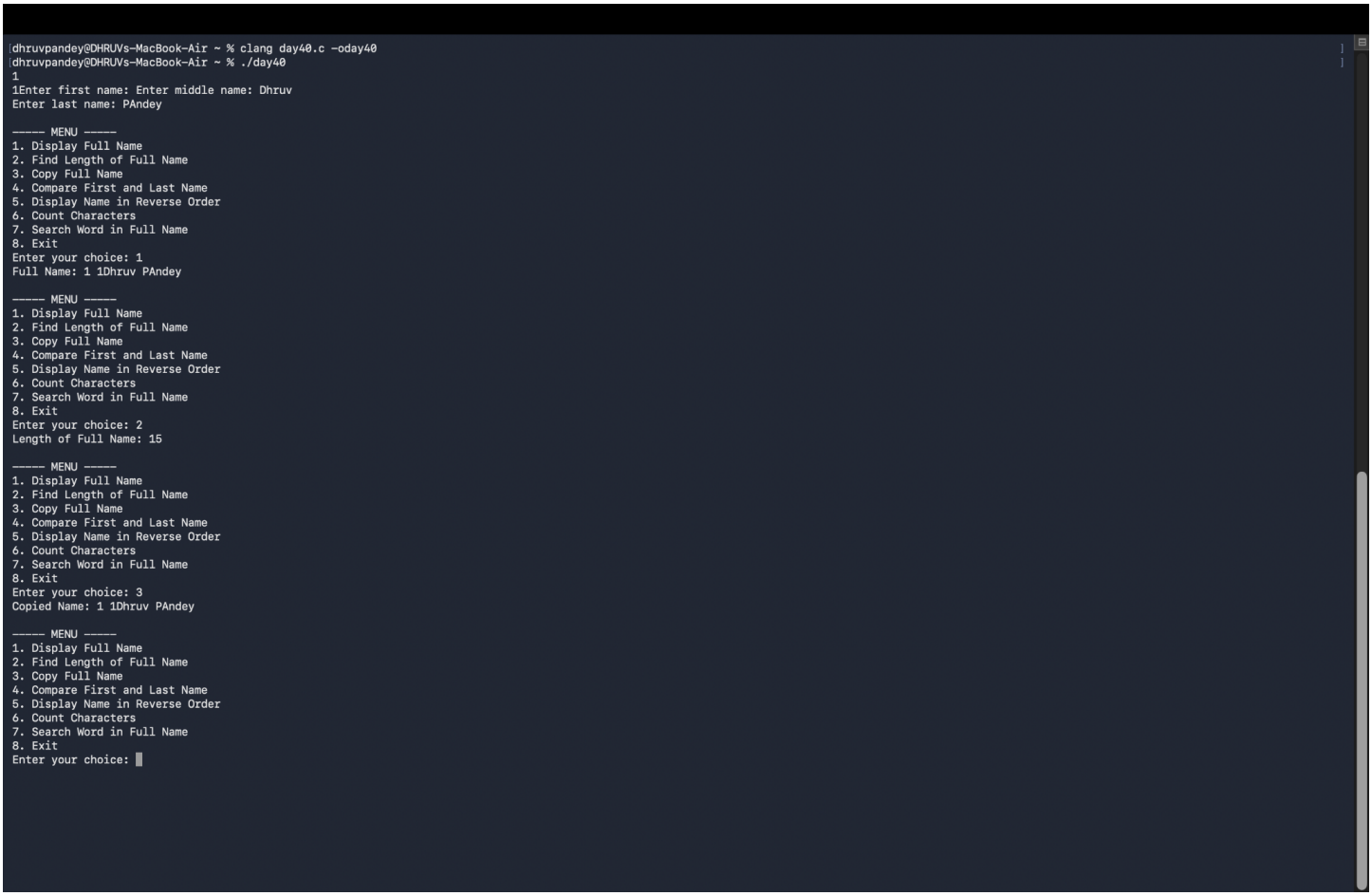
    do {
        printf("\n--- MENU ---\n");
        printf("1. Combine and display full name\n");
        printf("2. Find length of full name\n");
        printf("3. Copy full name\n");
        printf("4. Compare first and last name\n");
        printf("5. Display name in reverse order\n");
        printf("6. Count vowels, consonants, digits and spaces\n");
        printf("7. Search a word in full name\n");
        printf("8. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                combineNames(firstName, middleName, lastName, fullName);
                printf("Full name: %s\n", fullName);
                break;
            case 2:
                findLength(fullName, &length);
                printf("Length of full name: %zu\n", length);
                break;
            case 3:
                copyName(fullName, copiedName);
                printf("Copied name: %s\n", copiedName);
                break;
            case 4:
                compareNames(firstName, lastName, &comparisonResult);
                if (comparisonResult == 0)
                    printf("First name and last name are equal.\n");
                else
                    printf("First name and last name are different.\n");
                break;
            case 5:
                reverseNameOrder(firstName, middleName, lastName, reverseName);
                printf("Reverse order: %s\n", reverseName);
                break;
            case 6:
                countCharacters(fullName, &vowels, &consonants, &digits, &spaces);
                printf("Vowels: %d\n", vowels);
                printf("Consonants: %d\n", consonants);
                printf("Digits: %d\n", digits);
                printf("Spaces: %d\n", spaces);
                break;
            case 7:
                printf("Enter word to search: ");
                scanf("%49s", word);
                searchWord(fullName, word, &searchResult);
                if (searchResult)
                    printf("Word found in full name.\n");
                else
                    printf("Word not found in full name.\n");
                break;
            case 8:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 8);

    return 0;
}
```

Result

The program successfully performs all required string operations through a menu. It uses **stdio.h**, **string.h**, and **ctype.h**, with pointers for values returned or modified by functions.


dhruvpandey@DHRUVs-MacBook-Air ~ % clang day40.c -oday40
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day40
1
1Enter first name: Enter middle name: Dhruv
Enter last name: PAndey
----- MENU -----
1. Display Full Name
2. Find Length of Full Name
3. Copy Full Name
4. Compare First and Last Name
5. Display Name in Reverse Order
6. Count Characters
7. Search Word in Full Name
8. Exit
Enter your choice: 1
Full Name: 1 1Dhruv PAndey
----- MENU -----
1. Display Full Name
2. Find Length of Full Name
3. Copy Full Name
4. Compare First and Last Name
5. Display Name in Reverse Order
6. Count Characters
7. Search Word in Full Name
8. Exit
Enter your choice: 2
Length of Full Name: 15
----- MENU -----
1. Display Full Name
2. Find Length of Full Name
3. Copy Full Name
4. Compare First and Last Name
5. Display Name in Reverse Order
6. Count Characters
7. Search Word in Full Name
8. Exit
Enter your choice: 3
Copied Name: 1 1Dhruv PAndey
----- MENU -----
1. Display Full Name
2. Find Length of Full Name
3. Copy Full Name
4. Compare First and Last Name
5. Display Name in Reverse Order
6. Count Characters
7. Search Word in Full Name
8. Exit
Enter your choice: